Stat 420 Final Project - NuScenes Cut-In Risk Analysis

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Introduction

Understanding how nearby vehicles move relative to an autonomous vehicle is useful for analyzing traffic behavior. In this project, I use a small subset of the NuScenes autonomous driving dataset to study how a vehicle's current position, speed, and surrounding traffic relate to its lateral position five seconds later. This provides a simple way to describe short-term motion patterns without attempting to model the full complexity of cut-in behavior.

NuScenes includes detailed annotations of objects in urban driving scenes. For this project, I extracted ego-relative positions and velocities so that each observation represents a vehicle near the ego car at a given time. Only vehicles (not pedestrians or miscellaneous objects) were included to keep the dataset focused and easier to analyze.

The response variable is the vehicle's lateral position at time +5 seconds, and the predictors include several quantitative motion variables, traffic density, and a categorical indicator for vehicle type. These variables meet the course requirements for fitting a multiple linear regression model with both numerical and categorical predictors.

The main goal is simply to fit and evaluate a linear regression model, interpret which predictors appear most related to the future lateral position, and check basic regression assumptions. This analysis is intended to be exploratory rather than a full model of driving behavior.

Methods

Data Cleaning

This analysis uses a cleaned subset of NuScenes annotations. Each row represents a single vehicle at the moment it first enters a region where a lateral maneuver toward the ego lane is possible. All positions and velocities were converted into the ego reference frame so that distance and motion are measured relative to the autonomous vehicle.

```r
nusc <- read.csv("C:/nuScenes/clean_cutin_dataset.csv")

# View basic structure
str(nusc)
```

```
## 'data.frame':    409 obs. of  8 variables:
##  $ scene_id        : int  0 1 3 3 3 4 4 6 8 10 ...
##  $ instance_token  : chr  "b3bd430657754a16b32c7674507dba80" "25cce95097a544fc8bd3fb90205a79c8" "33a
##  $ distance_ahead  : num  25.86 2.61 34.14 36.99 34.91 ...
```

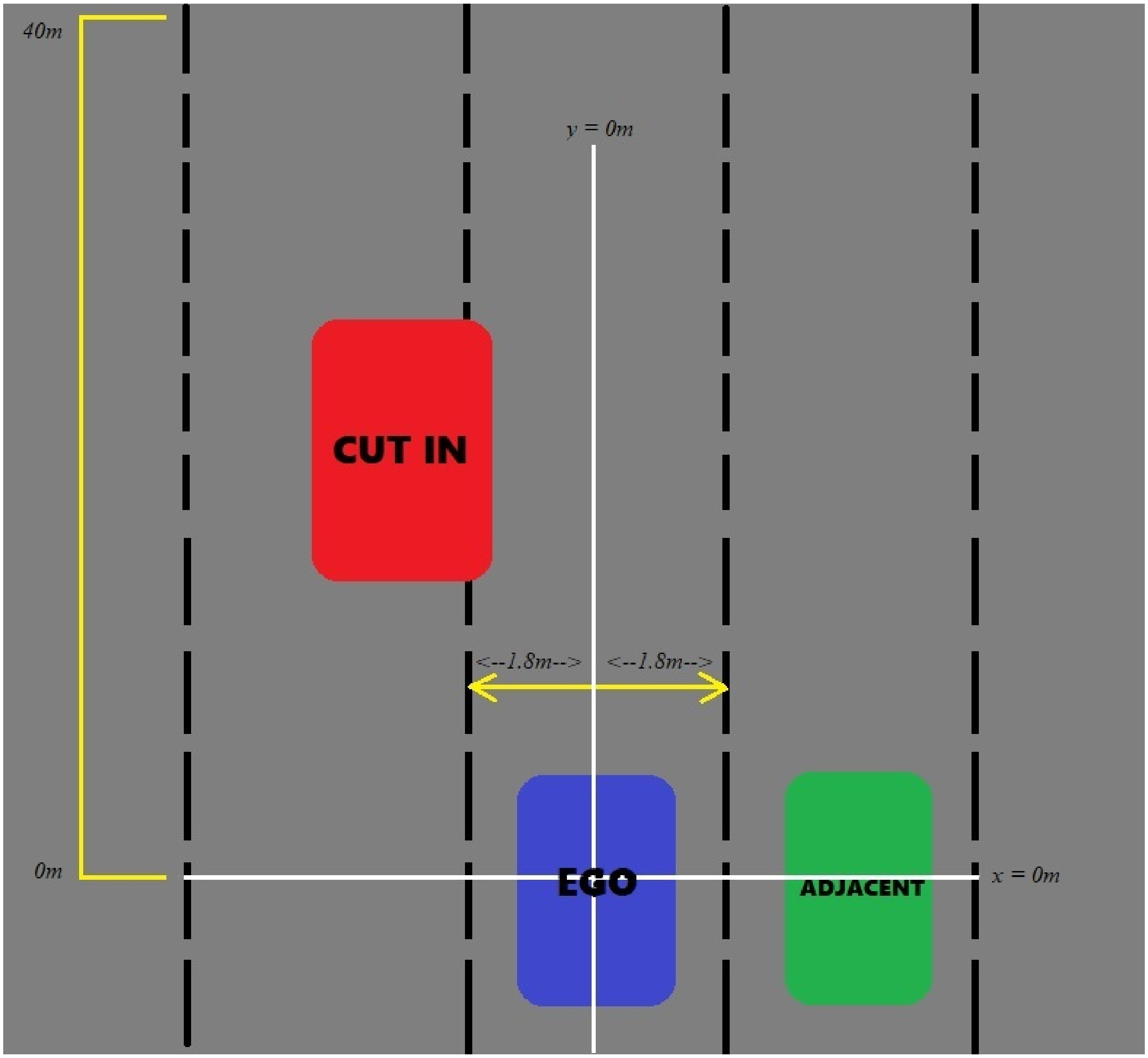


Figure 1: Cut-In Diagram

```
## $ lateral_offset  : num  6.85 -3.79 4.98 -4.19 -5.61 ...
## $ forward_velocity: num  0 0 0 0 0 ...
## $ side_velocity   : num  0 0 0 0 0 ...
## $ vehicle_type    : chr  "truck" "car" "car" "car" ...
## $ lateral_pos_5s  : num  4.61 -31.79 40.24 -4.82 -4.78 ...
```

The cleaned dataset contains 409 observations and 8 variables. Two identifier variables (scene_id and instance_token) were kept for reference but are not used as predictors. The remaining variables describe ego-relative geometry and motion:

- **distance_ahead:** Longitudinal (X-axis) offset from the ego vehicle.
- **lateral_offset:** Lateral (Y-axis) offset from the ego vehicle.
- **forward_velocity:** Ego-relative velocity along the X-axis.
- **side_velocity:** Ego-relative velocity along the Y-axis.
- **vehicle_type:** Categorical indicator of object class (car, truck, bus, motorcycle, bicycle).
- **lateral_pos_5s:** The same object's lateral position five seconds later.

All distances and velocities use the ego vehicle as the origin (0,0), which makes the variables easier to interpret in a driving context and more directly usable for regression modeling.

I checked all variables for missing values and incorrect data types. After preprocessing, no NA values remained, and each variable had an appropriate numeric or factor format.

Several filtering steps were applied to make the dataset more realistic for short-horizon motion modeling:

- Only vehicles were included; pedestrians, bicycles, and static objects were removed.
- Objects had to be moving generally forward (forward velocity > -1 m/s), so parked or oncoming vehicles were excluded.
- Only objects within 0–40 meters ahead of the ego vehicle were kept, since distant vehicles are unlikely to influence near-future lateral movement.

These steps produced a focused set of vehicle observations suitable for analyzing how present-time motion relates to lateral position five seconds later.

Exploratory Data Analysis

Cut-In-Like Outcomes As a simple first step, I checked how often vehicles ended up near the ego lane center five seconds later. Very few observations fell within a typical lane-width threshold, which suggests that strong lateral movements are uncommon in this subset. This is expected, since most vehicles in normal traffic do not dramatically shift lanes within such a short time window.

```r
sum(abs(nusc$lateral_pos_5s) <= 1.83) / nrow(nusc)
```

```
## [1] 0.04400978
```

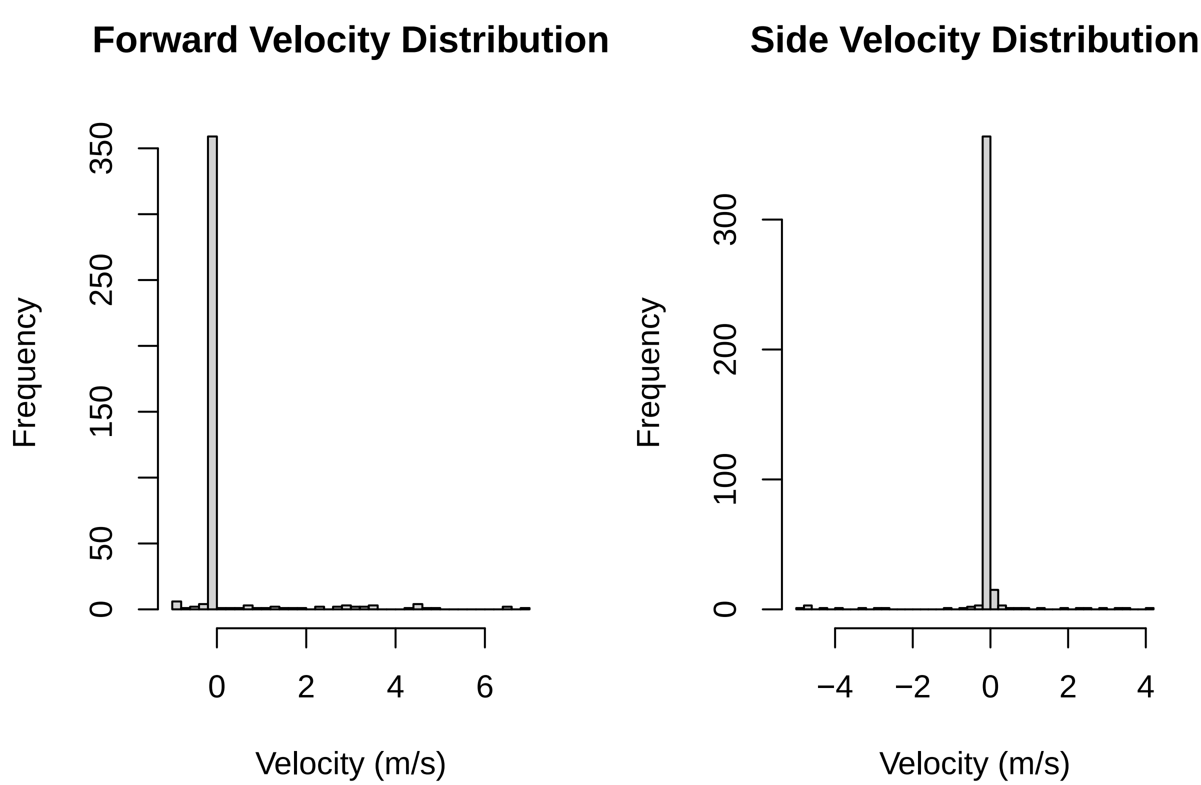
Vehicle Type Distribution I also looked at the distribution of vehicle types. Nearly all observations turned out to be cars, with other categories mostly filtered out during preprocessing. This imbalance reflects how the dataset was trimmed rather than any modeling decision, but it does mean the categorical variable will have limited variation in the final model.

```r
table(nusc$vehicle_type)
```

```
##
##     car trailer   truck
##     346      14      49
```

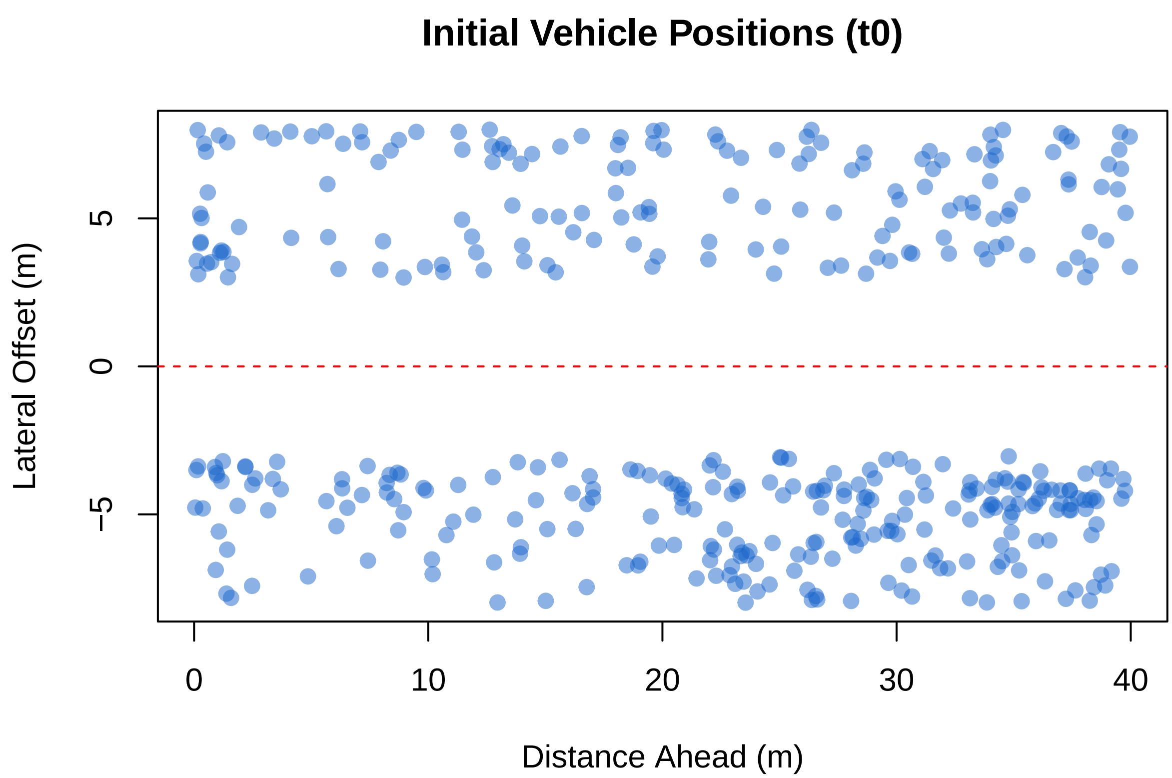
Velocity Histogram To get a sense of the motion characteristics, I plotted the distributions of forward and lateral velocities. Both are centered close to zero, which lines up with flowing traffic conditions—nearby vehicles tend to move at roughly the same speed as the ego vehicle. These patterns also confirm that the ego-relative velocity calculations behaved as expected.

```r
par(mfrow = c(1, 2))
hist(nusc$forward_velocity, breaks = 40,
     main = "Forward Velocity Distribution",
     xlab = "Velocity (m/s)")
hist(nusc$side_velocity, breaks = 40,
     main = "Side Velocity Distribution",
     xlab = "Velocity (m/s)")
```



Scatter Plot - Initial Positions I first looked at where vehicles were located when they entered the region of interest. As expected, almost all observations fall within the intended 0–40 meter forward range and within a reasonable lateral distance from the ego lane. This matches the filtering rules and suggests that the dataset correctly reflects vehicles that could potentially shift toward the ego lane.

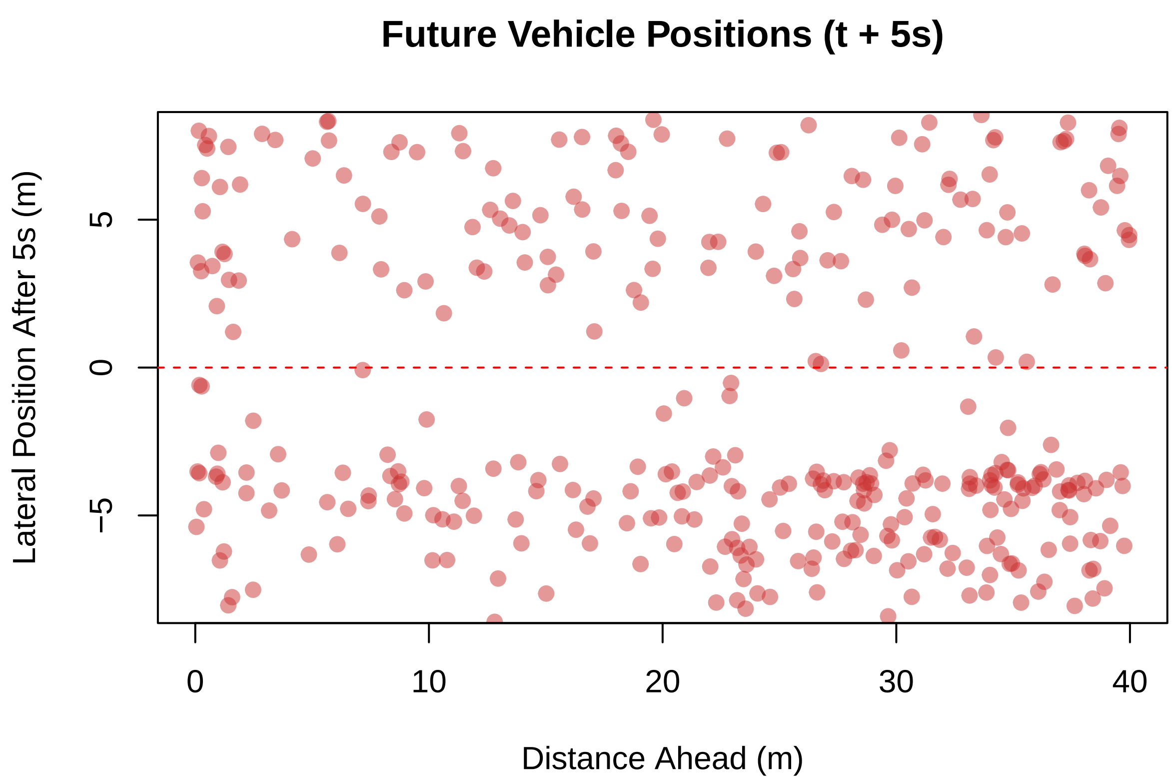
```r
# Initial positions (t0)
plot(
  nusc$distance_ahead, nusc$lateral_offset,
  main = "Initial Vehicle Positions (t0)",
  xlab = "Distance Ahead (m)",
  ylab = "Lateral Offset (m)",
  pch = 19, col = rgb(0.1, 0.4, 0.8, 0.5)
)
abline(h = 0, lty = 2, col = "red")  # Ego lane center
```



Scatter Plot - Postions T = +5 To see how these vehicles move over time, I plotted their lateral positions five seconds later. Most vehicles stay close to their original lateral placement, which matches earlier observations that strong lateral motion is uncommon in this sample. A small number move noticeably toward the ego lane center, giving a visual indication that some lane-changing behavior is present.

```r
# Positions 5 seconds later
# Use same limits as initial plot
xlim_range <- c(0, 40)
ylim_range <- c(-8, 8)
```

```r
plot(
  nusc$distance_ahead, nusc$lateral_pos_5s,
  main = "Future Vehicle Positions (t + 5s)",
  xlab = "Distance Ahead (m)",
  ylab = "Lateral Position After 5s (m)",
  pch = 19, col = rgb(0.8, 0.2, 0.2, 0.5),
  xlim = xlim_range,
  ylim = ylim_range
)
abline(h = 0, lty = 2, col = "red")
```



Correlation Matrix for Quantitative Predictors I computed a correlation matrix for the quantitative variables to check for obvious linear relationships. Most correlations are modest, which suggests multicollinearity is unlikely to be a major issue for the regression model. The one noticeably stronger relationship is between lateral_offset and the future lateral position, which matches the earlier observation that where a vehicle starts laterally is closely tied to where it ends up. The velocity variables show weaker associations with the response and with each other.

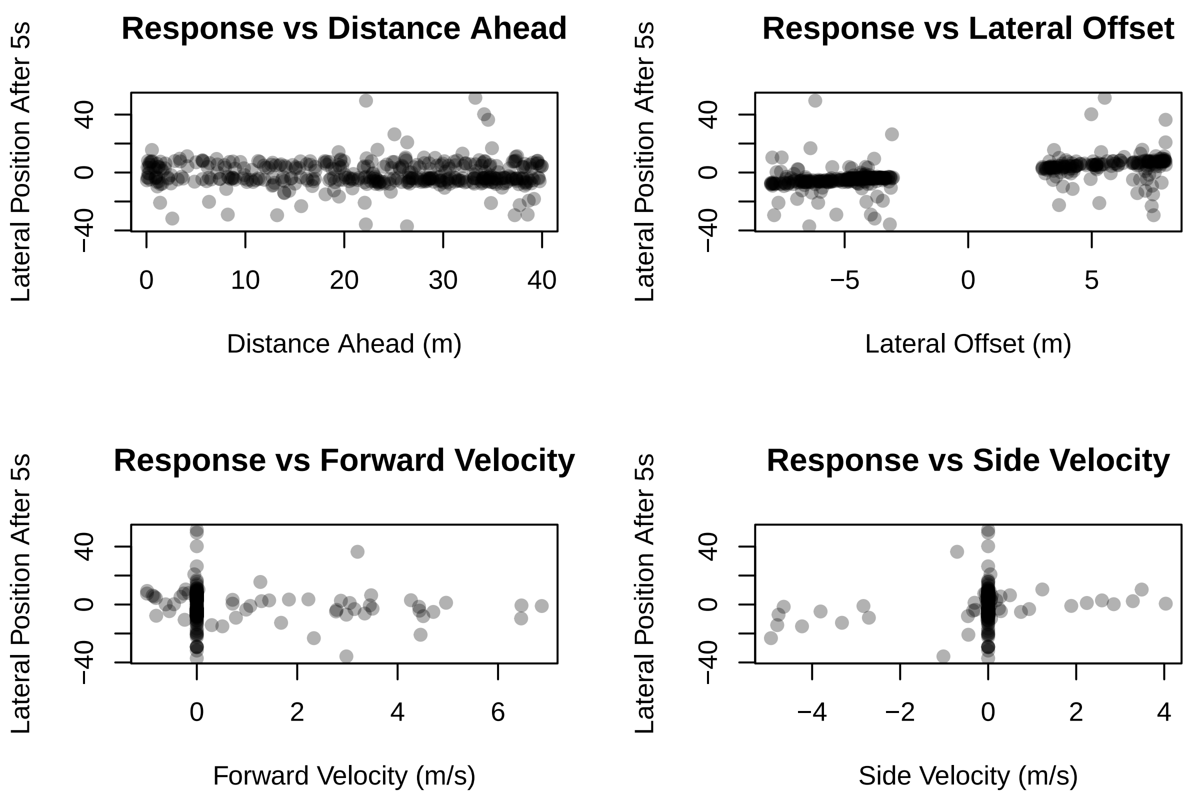
```r
quant_vars <- nusc[, c("distance_ahead","lateral_offset",
"forward_velocity","side_velocity","lateral_pos_5s")]
cor(quant_vars, use = "complete.obs")
```

```
##                  distance_ahead lateral_offset forward_velocity side_velocity
## distance_ahead       1.00000000    -0.13118188      -0.26158990    0.05838427
```

```
## lateral_offset      -0.13118188    1.00000000      0.08218649   -0.20018281
## forward_velocity    -0.26158990    0.08218649      1.00000000   -0.23330063
## side_velocity        0.05838427   -0.20018281     -0.23330063    1.00000000
## lateral_pos_5s      -0.02983251    0.54122884     -0.06454139    0.16486830
##                    lateral_pos_5s
## distance_ahead      -0.02983251
## lateral_offset       0.54122884
## forward_velocity    -0.06454139
## side_velocity        0.16486830
## lateral_pos_5s       1.00000000
```

Scatterplots: Response vs Quantitative Predictors I then looked at how the future lateral position relates to each quantitative predictor. These scatterplots help show whether any basic linear trends might be present. Most relationships appear quite weak, which is expected given that only a small fraction of vehicles make strong lateral movements. Lateral offset shows the most recognizable pattern, with vehicles generally remaining on the same side after five seconds. The velocity variables, on the other hand, form tight vertical clusters around zero and display a high amount of noise with no clear trend.

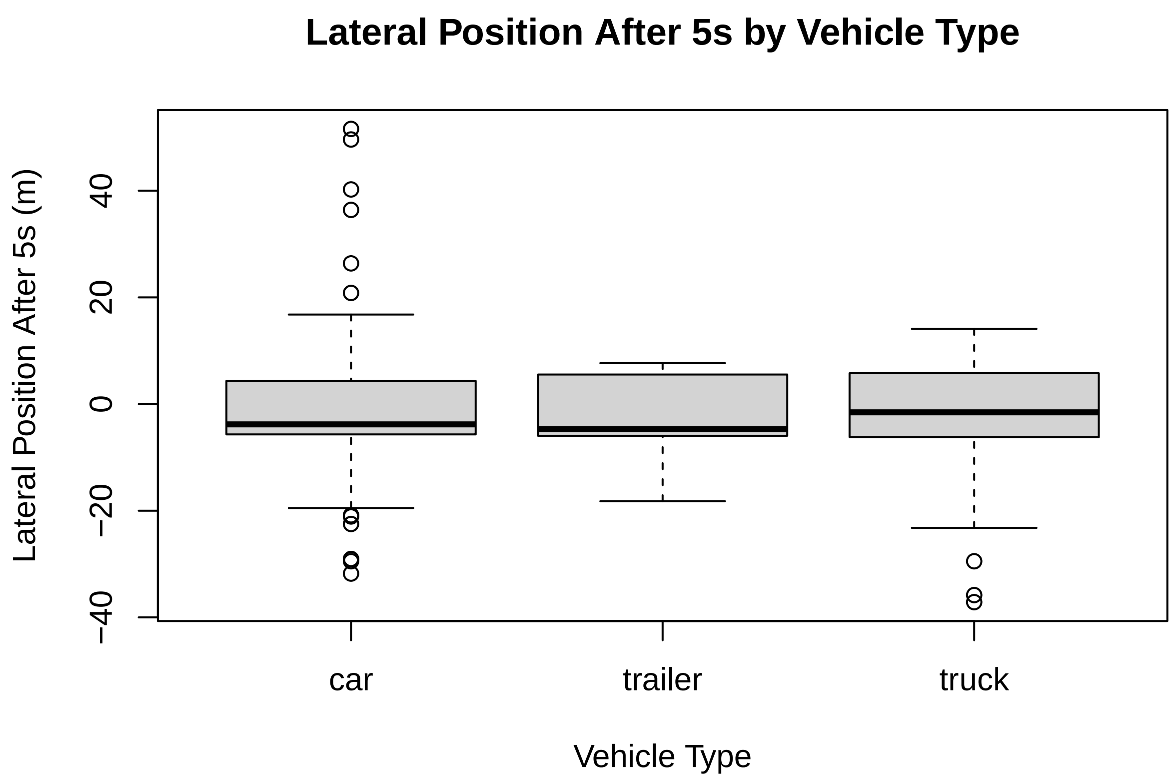
```r
par(mfrow = c(2, 2))

plot(nusc$distance_ahead, nusc$lateral_pos_5s,
xlab = "Distance Ahead (m)",
ylab = "Lateral Position After 5s",
main = "Response vs Distance Ahead",
pch = 19, col = rgb(0, 0, 0, 0.3))

plot(nusc$lateral_offset, nusc$lateral_pos_5s,
xlab = "Lateral Offset (m)",
ylab = "Lateral Position After 5s",
main = "Response vs Lateral Offset",
pch = 19, col = rgb(0, 0, 0, 0.3))

plot(nusc$forward_velocity, nusc$lateral_pos_5s,
xlab = "Forward Velocity (m/s)",
ylab = "Lateral Position After 5s",
main = "Response vs Forward Velocity",
pch = 19, col = rgb(0, 0, 0, 0.3))

plot(nusc$side_velocity, nusc$lateral_pos_5s,
xlab = "Side Velocity (m/s)",
ylab = "Lateral Position After 5s",
main = "Response vs Side Velocity",
pch = 19, col = rgb(0, 0, 0, 0.3))
```




```
par(mfrow = c(1, 1))
```

Boxplot: Response vs Vehicle Type Since vehicle_type is one of the predictors, I checked whether different classes show different patterns in future lateral position. The boxplot shows only small differences between the categories. Cars make up most of the dataset and therefore show the broadest spread, including several outliers. Trailers have a much narrower range, and trucks show a few more extreme values but otherwise look similar. Because of the imbalance in sample sizes, any categorical effects in the model should be interpreted cautiously.

```
boxplot(lateral_pos_5s ~ vehicle_type,
data = nusc,
main = "Lateral Position After 5s by Vehicle Type",
xlab = "Vehicle Type",
ylab = "Lateral Position After 5s (m)",
col = "lightgray")
```



EDA Final Thoughts Working with NuScenes required more preprocessing than a typical STAT 420 dataset. Many original attributes—such as vertical coordinates, non-vehicle objects, and vehicles far outside the region of interest—were not useful for this analysis and had to be removed. Several variables also needed to be converted into the ego-relative frame so that distances and motion could be interpreted in a driving context.

Because of this, the Data Cleaning and EDA steps are more detailed than in most course projects. The resulting dataset is much smaller and more focused, but still shows meaningful patterns: lateral offset has the clearest relationship with future position, while the velocity variables and distance ahead show weaker trends. These preprocessing steps were necessary to produce a coherent dataset suitable for the linear regression model that follows.

Initial Full Linear Model

For the first model, I fit a multiple linear regression using the future lateral position (lateral_pos_5s) as the response and all available motion variables plus vehicle_type as predictors.

```r
# Ensure vehicle_type is treated as a factor
nusc$vehicle_type <- factor(nusc$vehicle_type)

# Initial full linear model
full_model <- lm(lateral_pos_5s ~ distance_ahead +
                                  lateral_offset +
                                  forward_velocity +
                                  side_velocity +
```

```r
                                    vehicle_type, data = nusc)

summary(full_model)
```

```
##
## Call:
## lm(formula = lateral_pos_5s ~ distance_ahead + lateral_offset +
##     forward_velocity + side_velocity + vehicle_type, data = nusc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.006  -0.581   0.016   0.889  55.721
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          0.015669   0.836890   0.019  0.98507
## distance_ahead       0.006842   0.030968   0.221  0.82525
## lateral_offset       1.015661   0.066756  15.214  < 2e-16 ***
## forward_velocity    -0.458230   0.381061  -1.203  0.22987
## side_velocity        3.229529   0.512162   6.306 7.56e-10 ***
## vehicle_typetrailer -0.744340   1.960170  -0.380  0.70434
## vehicle_typetruck   -3.200130   1.123468  -2.848  0.00462 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.174 on 402 degrees of freedom
## Multiple R-squared:  0.3859, Adjusted R-squared:  0.3768
## F-statistic: 42.11 on 6 and 402 DF,  p-value: < 2.2e-16
```

Overall, the model explains a modest portion of the variation in the response, which is reasonable given how noisy short-term lateral movement is in this dataset.

Among the quantitative predictors, lateral_offset is by far the strongest. Its coefficient is close to one, meaning vehicles tend to end up roughly on the same side they started from. Side_velocity also shows a meaningful positive effect, indicating that vehicles already drifting sideways tend to continue in that direction. The other quantitative predictors—distance ahead and forward velocity—do not contribute much.

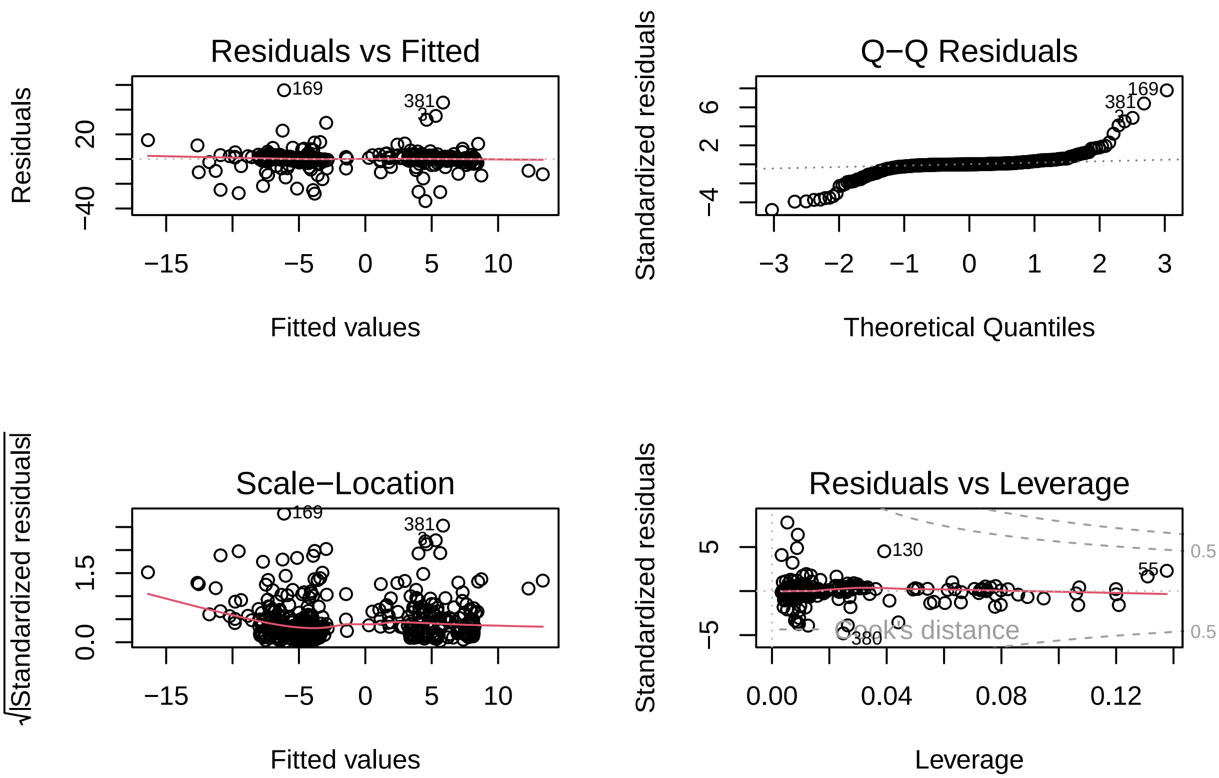
For the categorical variable, the model uses car as the baseline category. The coefficients for trailer and truck represent how those types differ from cars after accounting for the motion variables. Trailers do not differ much from cars, while trucks show a small negative shift. Because the dataset contains far more cars than the other categories, these comparisons should be interpreted cautiously.

In short, the full model mainly reflects what appeared in the EDA: starting lateral position has the clearest relationship with where a vehicle ends up five seconds later, with side-to-side motion providing a smaller secondary effect. The remaining predictors play relatively minor roles, but the model still serves as a reasonable baseline for further evaluation.

Diagnostics and Cross Validation

Residuals, Normality, Homoscedasticity After fitting the full model, I examined the standard diagnostic plots—residuals vs. fitted, the Q-Q plot, the scale-location plot, and the residuals vs. leverage plot—to evaluate whether the linear model assumptions appear reasonable.

```r
par(mfrow = c(2, 2))
plot(full_model)
```




```r
par(mfrow = c(1, 1))
```

The residuals vs. fitted plot shows a mostly scattered pattern around zero, which suggests that a linear form is not wildly inappropriate. However, there are a few points with noticeably large residuals, and the spread is not perfectly uniform. For an exploratory model on noisy driving data, this level of deviation is expected and not severe enough to prevent using the model.

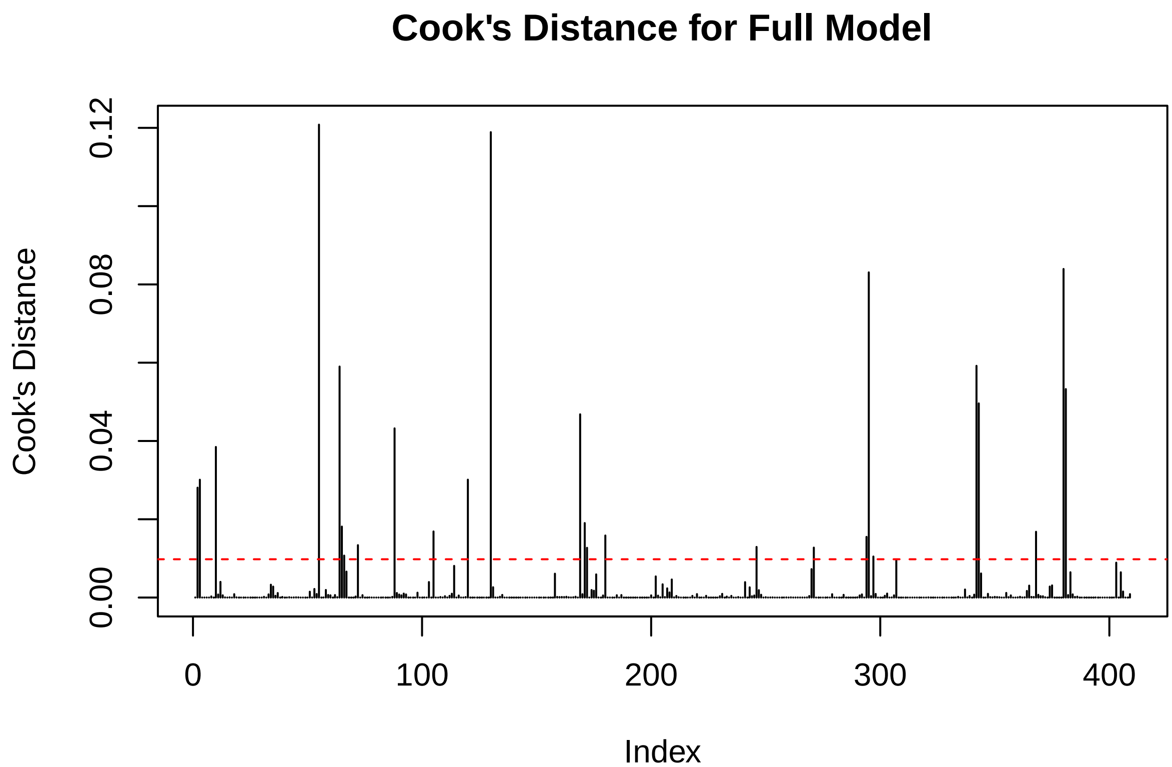
The Q-Q plot shows clear deviations from the 45-degree line, especially in the upper tail. The residuals bend upward, indicating heavier-than-normal positive residuals. This matches what we would expect, since a small number of vehicles make unusually large lateral movements. Normality is not perfect, but the results are still interpretable for a basic linear model.

The scale-location plot indicates some heteroscedasticity: the spread of residuals decreases slightly as fitted values increase. The pattern is not dramatic, but it does show that variance is not completely constant across the range of predictions. For this dataset, some non-constant variance is expected and does not invalidate the basic conclusions.

The residuals vs. leverage plot shows several points with high standardized residuals but relatively low leverage, meaning they affect model fit mainly through their extreme response values rather than unusual predictor combinations. One or two points approach the Cook's distance contours, but none appear to dominate the model. This is consistent with the presence of a few genuine outlier lateral movements.

Influential Observations I also checked Cook's distance to identify any observations that might have a large influence on the fitted model. This helps show whether a small number of points are disproportionately affecting the regression results.

```
plot(cooks.distance(full_model), type = "h",
     main = "Cook's Distance for Full Model",
     ylab = "Cook's Distance")
abline(h = 4 / nrow(nusc), col = "red", lty = 2)
```



The horizontal line gives a common rule-of-thumb cutoff for potentially influential points. In this dataset, a few observations fall at or above this level. This is not surprising, since a small number of vehicles exhibit larger lateral movements than the rest. These points were kept in the analysis because they reflect real behavior in the data rather than errors or anomalies.

Diagnostics Final Thoughts Overall, the diagnostic plots suggest that the linear model is reasonable for an exploratory analysis, but not perfect. There is some mild nonlinearity, a few larger residuals, and signs of non-constant variance. These issues are expected given the noisy nature of short-term lateral movement and the presence of a few genuine outlier behaviors. While there are ways the model could potentially be improved—such as transformations or adding interactions—the current model is adequate for the purposes of this project.

Additional Modeling Methods and Improvements

Motivation The diagnostic plots from the full model suggest a few areas where the fit might be improved. There are mild signs of nonlinearity, some non-constant variance, and a handful of observations with unusually large residuals. These issues are not severe, but they indicate that a slightly more flexible or simplified model might describe the data better. To explore this, I considered a few common extensions to the initial model, including adding interaction terms, applying simple transformations, and using AIC-based stepwise selection to see whether a more concise model would perform similarly.

Interaction Terms To check whether the effect of initial lateral position depends on how a vehicle is moving, I fit a model that included interactions between lateral_offset and both velocity variables. These terms allow the influence of starting lateral position to vary depending on whether the vehicle is drifting sideways or moving forward.

```r
interaction_model <- lm(lateral_pos_5s ~ distance_ahead +
                                         lateral_offset +
                                         forward_velocity +
                                         side_velocity +
                                         vehicle_type +
                                         lateral_offset:forward_velocity +
                                         lateral_offset:side_velocity,
                        data = nusc)

summary(interaction_model)
```

```
##
## Call:
## lm(formula = lateral_pos_5s ~ distance_ahead + lateral_offset +
##     forward_velocity + side_velocity + vehicle_type + lateral_offset:forward_velocity +
##     lateral_offset:side_velocity, data = nusc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.983  -0.581  -0.005   0.867  55.701
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      0.030225   0.839671   0.036  0.97130
## distance_ahead                   0.005844   0.031157   0.188  0.85132
## lateral_offset                   1.011114   0.068316  14.800  < 2e-16 ***
## forward_velocity                -0.486236   0.388842  -1.250  0.21186
## side_velocity                    3.283212   0.560005   5.863 9.53e-09 ***
## vehicle_typetrailer             -0.759081   1.966167  -0.386  0.69965
## vehicle_typetruck               -3.191049   1.140546  -2.798  0.00539 **
## lateral_offset:forward_velocity  0.027064   0.079185   0.342  0.73269
## lateral_offset:side_velocity    -0.006917   0.079656  -0.087  0.93085
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.19 on 400 degrees of freedom
## Multiple R-squared:  0.3862, Adjusted R-squared:  0.3739
## F-statistic: 31.45 on 8 and 400 DF,  p-value: < 2.2e-16
```

In this expanded model, neither interaction term is statistically significant, and their coefficients are both very close to zero. This indicates that the relationship between where a vehicle starts laterally and where it ends up five seconds later does not meaningfully depend on its forward or sideways velocity. The main effects from the full model remain dominant: lateral_offset is still by far the strongest predictor, and side_velocity continues to show a smaller but meaningful effect.

The overall fit of the interaction model is nearly identical to the original model, with only minor changes in residual variation and R-squared. Because the additional terms do not improve interpretability or predictive performance, they do not appear to add useful structure to the model.

Transformations (obrief exploration) Because the scatterplot of lateral_offset versus the response showed a small amount of spread that could possibly suggest curvature, I fit a model that included a squared term for lateral_offset to test for a nonlinear effect.

```r
quad_model <- lm(lateral_pos_5s ~ distance_ahead +
                                  lateral_offset +
                                  I(lateral_offset^2) +
                                  forward_velocity +
                                  side_velocity +
                                  vehicle_type, data = nusc)

summary(quad_model)
```

```
##
## Call:
## lm(formula = lateral_pos_5s ~ distance_ahead + lateral_offset +
##     I(lateral_offset^2) + forward_velocity + side_velocity +
##     vehicle_type, data = nusc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.191  -0.571   0.042   0.938  55.612
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -0.302889   1.052855  -0.288  0.77374
## distance_ahead       0.006318   0.031014   0.204  0.83868
## lateral_offset       1.009961   0.067786  14.899  < 2e-16 ***
## I(lateral_offset^2)  0.010548   0.021119   0.499  0.61773
## forward_velocity    -0.438663   0.383424  -1.144  0.25328
## side_velocity        3.240139   0.513080   6.315 7.18e-10 ***
## vehicle_typetrailer -0.758002   1.962193  -0.386  0.69948
## vehicle_typetruck   -3.241501   1.127565  -2.875  0.00426 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.18 on 401 degrees of freedom
## Multiple R-squared:  0.3863, Adjusted R-squared:  0.3756
## F-statistic: 36.06 on 7 and 401 DF,  p-value: < 2.2e-16
```

In this model, the squared term is not statistically significant, and its coefficient is extremely small. The overall model fit — including R-squared, adjusted R-squared, and residual standard error — is nearly identical to the original linear model. This indicates that adding curvature does not meaningfully improve the model or change the interpretation of the predictors.

Because the quadratic term provides no practical benefit, the simpler linear relationship is retained in the final model.

Stepwise AIC Model Selection To see whether a simpler model might perform as well as the full model, I applied AIC-based stepwise selection starting from the full set of predictors.

```r
library(MASS)
step_model <- stepAIC(full_model, direction = "both", trace = FALSE)
summary(step_model)
```

```
##
## Call:
## lm(formula = lateral_pos_5s ~ lateral_offset + side_velocity +
##     vehicle_type, data = nusc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.928  -0.543   0.036   0.923  55.797
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          0.06159    0.39224   0.157  0.87531
## lateral_offset       1.01071    0.06632  15.239  < 2e-16 ***
## side_velocity        3.37399    0.49882   6.764 4.72e-11 ***
## vehicle_typetrailer -0.75958    1.95570  -0.388  0.69793
## vehicle_typetruck   -3.19713    1.11689  -2.863  0.00442 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.171 on 404 degrees of freedom
## Multiple R-squared:  0.3833, Adjusted R-squared:  0.3772
## F-statistic: 62.77 on 4 and 404 DF,  p-value: < 2.2e-16
```

The stepwise procedure selected a reduced model containing only lateral_offset, side_velocity, and vehicle_type. Both distance_ahead and forward_velocity were removed, indicating that they contribute very little once the other predictors are included.

The reduced model has nearly the same residual standard error and R-squared as the full model, meaning the dropped variables provided minimal additional explanatory power. The two strongest predictors remain the same as before: lateral_offset has the largest effect, and side_velocity provides a smaller but meaningful adjustment. The categorical differences for vehicle type also persist.

Overall, the AIC-selected model is simpler but retains the key structure of the original model, suggesting that the removed variables were not essential for describing short-term lateral movement in this dataset.

ANOVA Model Comparison To formally evaluate whether the interaction model or the quadratic model provide any improvement over the full model, I compared the nested models using ANOVA.

```r
anova(full_model, interaction_model)
```

```
## Analysis of Variance Table
##
## Model 1: lateral_pos_5s ~ distance_ahead + lateral_offset + forward_velocity +
##     side_velocity + vehicle_type
## Model 2: lateral_pos_5s ~ distance_ahead + lateral_offset + forward_velocity +
##     side_velocity + vehicle_type + lateral_offset:forward_velocity +
```

```
##     lateral_offset:side_velocity
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1    402 20688
## 2    400 20680  2    7.7483 0.0749 0.9278
```

```r
anova(full_model, quad_model)
```

```
## Analysis of Variance Table
##
## Model 1: lateral_pos_5s ~ distance_ahead + lateral_offset + forward_velocity +
##     side_velocity + vehicle_type
## Model 2: lateral_pos_5s ~ distance_ahead + lateral_offset + I(lateral_offset^2) +
##     forward_velocity + side_velocity + vehicle_type
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1    402 20688
## 2    401 20675  1    12.861 0.2495 0.6177
```

The ANOVA comparisons show that neither the interaction model nor the quadratic model provides a meaningful improvement over the full model. For both tests, the reduction in residual sum of squares is extremely small, and the p-values are large, indicating no statistically significant gain in fit. This suggests that the added complexity in these expanded models is not justified for this dataset.

Conclusion

This project applied a linear regression model to a heavily filtered subset of NuScenes data in order to study lateral vehicle movement over a short five-second horizon. Because the dataset required extensive pruning to isolate vehicles that could realistically attempt a lateral maneuver, most of the exploratory analysis ended up confirming the constraints already imposed by the cleaning process. Strong lateral movements were rare, most vehicles stayed in their original lane region, and vehicle type showed little variation after filtering.

The modeling results were straightforward. The strongest predictors of where a vehicle ended up after five seconds were its starting lateral position and, to a lesser extent, its sideways velocity at the initial observation. These findings make intuitive sense: in the narrow conditions examined here, lane-changing behavior is almost entirely determined by whether a vehicle has already begun moving sideways. Other predictors, including distance ahead, forward velocity, and most vehicle-type effects, contributed very little once these two variables were included.

Attempts to improve the model with interaction terms, nonlinear transformations, and AIC-based selection did not reveal any additional structure. The more complex models performed nearly the same as the initial linear model, which suggests that the relationship captured in this dataset is essentially linear and limited. In other words, this analysis reduces to a simple statement: when lateral movement occurs, it is already visible in the starting lateral offset and sideways drift. Nothing else in this constrained subset meaningfully changes the outcome.

Although this limits the statistical depth of the project, it also reflects the reality of the data and the modeling choices made for a STAT 420 assignment. The point here is not to force an impressive result from a situation where little signal exists, but to demonstrate an understanding of how to evaluate a model, interpret its predictors, check assumptions, and verify whether added complexity is justified. From that standpoint, the analysis behaved exactly as expected.

It is also worth noting that while this small-scale analysis is not directly useful in real-world autonomous-vehicle systems, a broader version of the same idea can play an important role. In practice, short-horizon lateral behavior would be modeled using richer, real-time perception data and methods suited to nonlinear, dynamic environments. Such models can contribute to components of decision-making or planning cost functions by estimating how likely nearby agents are to move toward the ego vehicle. That type of macro-level modeling lies well beyond the scope of a linear regression assignment, but the high-level reasoning about which signals matter is consistent with how AV systems approach risk and intent prediction.

Overall, this project shows that even when the statistical model ends up being simple and the findings limited, the process of cleaning, exploring, modeling, and evaluating still demonstrates a solid understanding of the methods taught in the course.